Chapter 34 The Electric Charge
Calculus Physics II

All interactions (that we know of) in the universe are produced by the four fundamental forces of nature and they are:

Gravitational force – created by the mass of an object.
Electromagnetic Force (Electric and magnetic force) – Created by charge of the proton and electron.
Weak force – force in the nucleus of the atom responsible for radioactive decay.
Strong force - keeps the protons and neutrons stuck together in the nucleus.

- In Physics I we studied the Gravitational force; in Physics II we will study Electric and magnetic forces and how they are related.
- All of these Fundamental forces are non-contact forces and interact through the fields they generate.

Fields (or Force Fields) - In physics a Field exists in a region of space if an object placed in that region and the object experiences a force due to some quantity that object processes (mass, charge, magnetism, etc. …). The field can exist in empty space and creates non-contact forces, such as the force of the sun on the earth or an apple falling to earth (gravitational field) or proton on the electron (electric Field).

- With non-contact forces the force acts through its field.
- Fields are real things. Fields have energy, fields are energy!
- Fields are conditions in space that produce forces.
- Fields are generated by some intrinsic property of the object (mass, charge, etc. …).

Electromagnetism - The combined effect of the Electric and the Magnetic forces and fields.

Electrostatics – the study of the electric charge at rest, the charge is not moving.

The word Electricity is derived from the Greek word for amber, "elektron". It was the ancient Greeks who first made the observation that after amber (fossilized tree resin) was rubbed with a cloth it would attract small pieces of lint. This is now known as the electrostatic force.



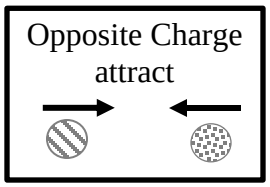
Examples of the electrostatics: static cling, Balloon on wall, hair standing up after combing, lightning, shocked by doorknob.

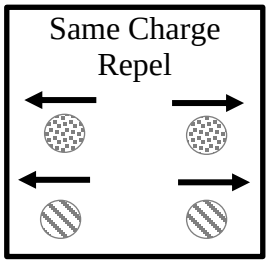
Electric Charge (q) – the Electric Charge is an intrinsic property of the proton and electron.

Units of charge are **Coulombs (C)** which is the SI unit for charge.

- There are 6,250,000,000,000,000,000 electrons in one coulomb of charge
- Most common charges encountered in everyday life are on the magnitude of 10^{-9} C or 1 nC (nano-coulomb).

- There are only two kinds of charge and they are called **positive (+) & negative (-).**
- **"Positive" and "negative" are only names and not algebraic signs!!**
- Like charges repel and unlike charges attract.
- The source of charge is from the electron and proton.
- When positive charge is produced an equal amount of negative charge is also produced.
- Charge can be transferred from one object to another.
- A net charge creates a force on other objects with a net charge.





How to think about charge – The quantity that is called "charge" comes from the electron and from the proton. It is called negative (electron charge) and positive (proton charge) but there is nothing negative or positive about the charge! The positive or negative does not make the charge algebraically "-" or "+", the charges could have just as easily been called "green" charge and "blue" charge. Green charge attracts Blue charge and green charge is repelled by other green charge, blue charge is repelled by other blue charge and is attracted to green charge. It is better to think of the charge as the "Electron Charge" and the "Proton Charge". There is nothing "negative" about the electron's charge and there is nothing "positive" about the protons charge. It is just a name or label give to the particular charge.

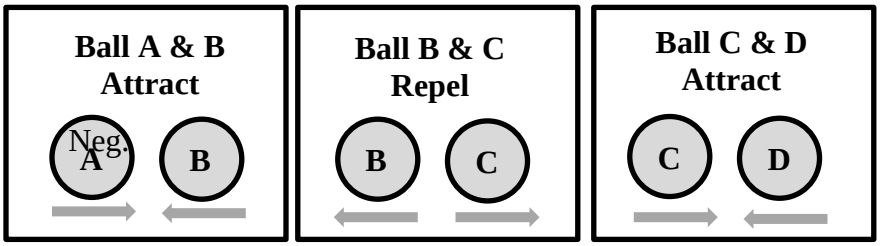
Net Charge – when there is an excess of positive or negative charge on an object.

- By having more electron charge (-) than proton charge (+) the object has a net negative charge.
- By having more proton charge (+) than electron charge (-) the object has a net positive charge.

Neutral charge – equal number of positive and negative charge, zero net charge.

- Neutral does NOT mean there is no charge.

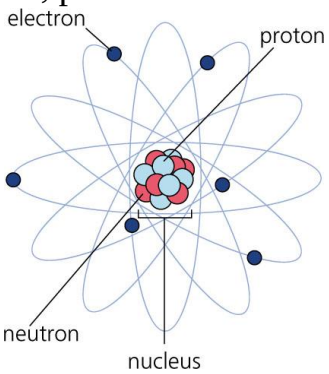
Example of Charge interaction: For the 4 balls that have a net charge shown below, Ball A, Ball B, Ball C and Ball D. If Ball A has a negative charge, given the data below, what are the net charges on Ball's B, C and D?



- ❖ Because Ball A is attracted to Ball B and Ball A has a negative charge that means that Ball B must have a positive charge.
- ❖ Because Ball B and Ball C repel, and Ball B has a positive charge that means that Ball C must have a Positive charge because it is repelled by Ball B.
- ❖ Because Ball C and Ball D attract, and Ball C has a positive charge that means Ball D must have a negative charge because it is attracted to Ball C.

Ball A – negative net charge
Ball B – positive net charge
Ball C – positive net charge
Ball D – negative net charge

The Atom - The atom is the smallest particle of an element that can be identified as that element. The atom is made up of three fundamental particles: the electron, proton and neutron.



- If there is an equal number of positive protons as there are negative electrons in the atom for an object that is neutral (has no net charge).
- If there are an **unequal** number of protons and electrons, then the atom is charged.
- Most of the mass of the atom is in the nucleus.

Particle	Charge	Mass
Electron	$q_e = 1.6 \times 10^{-19}$ C negative	$m_e = 9.11 \times 10^{-31}$ kg
Proton	$q_p = 1.6 \times 10^{-19}$ C positive	$m_p = 1.673 \times 10^{-27}$ kg
Neutron	Neutral q = 0 C no net charge	$m_n = 1.675 \times 10^{-27}$ kg

- ❖ The charge cannot be separated from the electron or proton, it is an intrinsic property of the particle.
- ❖ The magnitude of the charge on the proton is exactly equal to the magnitude of the charge on the electron, they only have different type of charge.

Proton (p) – is a subatomic particle that is part of the nucleus of the atom, has a positive charge and it has slightly less mass than the neutron.

Charge on the proton (q_p or p or p^+) $q_p = 1.6 \times 10^{-19}$ C and is called positive

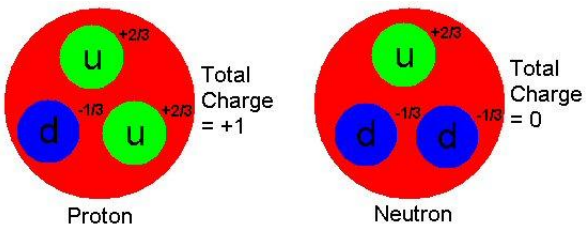
- The proton defines what element the atom is and its chemical properties.
- The atomic number (Z) gives the number of protons in the nucleus of the atom.
- All protons are exactly the same (mass and charge).
- The diameter of the proton is about 10^{-15} m or 1 femtometer (fm).
- The proton is made up of smaller particles known as quarks.
- The proton interacts with other protons and neutrons through the strong force. The Strong Force is responsible for keeping the protons and neutrons glued together inside the Atom.

Neutron (n) – is a subatomic particle that is part of the nucleus of the atom, has no net charge and has slightly more mass than the proton.

- All neutrons are exactly the same (mass and no net charge).
- The neutron is also made up of quarks.
- Because the neutron does not have a charge it acts to stabilize the nucleus.
- The neutron interacts with other neutrons and protons through the strong force.

Quarks – A Quark is a fundamental particle that makes up the protons and neutrons. There are two types of Quarks, the up quark and the down quark. There are other types of quarks (strange, charm, top and bottom quarks) and another particle made from them. The quark is not known to be made of any smaller particle and is pure energy.

- The Up quark has a charge of positive 2/3.
- The Down quark has a charge of negative 1/3.
- Two Up and a Down quark make a proton.
- Two Down and an Up quark make a neutron.



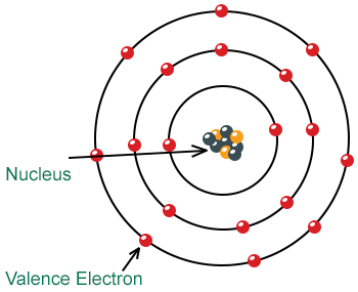
- Quarks can NOT exist by themselves therefore there can never be a fraction of a charge alone in nature.

Electron (e) – is a subatomic particle that has a very small mass relative to the proton. The electron has a charge that is equal in magnitude to the proton but opposite in the type of charge of the proton, the electron resides outside the nucleus of the atom.

Charge on the electron (q_e or e^- or q^-) $q_e = 1.6 \times 10^{-19}$ C and is called negative

- The electron's mass is about 2,000 times less massive than the protons mass.
- The upper limit on the diameter of the electron is about 10^{-18} meters.
- All electrons are exactly the same (mass and charge).
- The electrostatic force is what keeps the electrons bound to the atom.
- For an atom to be neutral there must be as many electrons (negative charge) as protons (positive charge).
- Scientists believe that the electron is a fundamental particle and is not made up of any smaller particles.

Valence electrons – are the outer most electrons in an atom. The outermost electrons are responsible for how atoms and molecules will react with each other and their environment. Valence electrons determine the properties and chemical reactivity of the element and participate in chemical bonding.



Electrical Insulators and conductors – In the study of electromagnetism, materials are broadly classified based on their ability to allow the flow of electric charge. In materials it is the electrons that are able to move not the protons, the protons are bound in the nucleus and can only move with the atom or molecule.

Insulator and Conductors depends on the material and its properties and the environment. Examples would be pure water which is a poor conductor (insulator) but salt water is a good conductor. Ceramic is a poor conductor but ceramic at a very low temperature is very good conductor.

Insulator (Dielectric) – poor conductor of electric charge, limited movement of electrons. The electrons are tightly bound to their atoms and cannot move freely. Valence electrons in atoms are tightly bound. Materials that resist the flow of electric charge.

Examples are plastic, wood, rubber, glass, pure water

Applications: Coating for wires and cables, dielectric materials in capacitors, safety equipment (e.g., gloves, mats).

Conductor – a material that electric charge can easily flow, abundant supply of free electrons the valence electrons are not tightly bound and are free to move forming a "sea of electrons.". The electrons can also redistribute charge quickly when exposed to external electric fields.

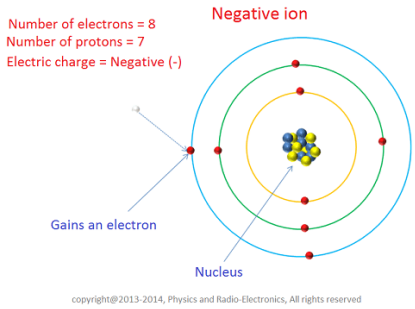
Examples: Metals like copper, silver, aluminum, and gold.

Applications: Wiring and electrical circuits, grounding systems, electrostatic shielding.

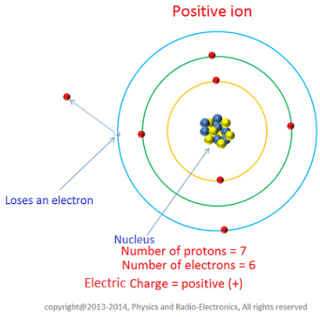
Semiconductor – Insulating material that an impurity is added to change its electrical properties and can be changed from an insulator to a conductor by changing some eternal property. Materials like silicon and germanium that have conductivity between conductors and insulators. Their behavior can be modified by doping and external conditions (temperature, light, etc.).

The Ion – an atom with a net charge due to losing or gaining electrons.

Positive Ion – when an electron is removed from a neutral atom, this gives the atom a net positive charge and the separated electron is a free negative charge.



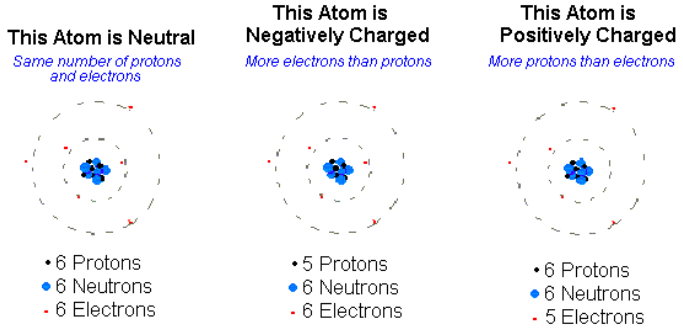
Negative Ion – when a neutral atom gains a negatively charged electron.



Ionization – adding or removing an electron from the atom leaving a net charge on the atom. Turning the atom into an Ion. For an object with a net charge the number of protons N_p does not equal the number of electrons N_e then the total charge Q of an object is given by

$$Q_{ion} = (N_p - N_e)q_e$$

- Net Positive charge – electrons have been removed from a neutral object
- Net Negative charge – electrons have been add to a neutral object
- The number of protons is set for the atom and does not change it is the electrons that are added or removed.

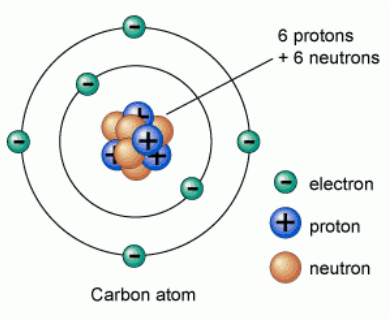


- Because electrons and protons have mass, adding charge increase the mass and removing charge decreases the mass of the object being charged.

Charge and the Elements – The charge of an element is neutral unless electrons have been removed or added. The Periodic table of the elements gives the number of protons in the nucleus (Atomic number Z) and the mass of one mole of the element (Atomic Mass u in grams/mole). The number of protons defines the element and the charge of the nucleus $Q_{nucleus}$.

$$Q_{nucleus} = (\text{Atomic \#})(\text{charge on proton}) = Zq_p$$

- ❖ There must be the same number of electrons than there are protons in order to have a neutral element therefore the charge of the electrons surrounding the nucleus must be the same as the charge of the nucleus.
- ✓ For the carbon atom (Z = 6) with 6 protons, in order for the atom to be neutral it must have 6 electrons.



Conservation (in Physics) - A quantity is conserved when the total quantity does not change over time, it only moves around to different places or different forms. Examples are: Energy, momentum, mass and now charge.

Conservation of charge – The electric charges can be produced only in pairs of equally strong single proton charge (positive) and single electron charge (negative) charges. The net charge of the universe never changes.

- The total charge in a closed system does not change over time.
- Charge is neither created nor destroyed it is transferred from one object to another.

Charge Quantization – All charge is a multiple of the fundamental charge (electron and proton charge), all transfer of charge is a multiple of this fundamental charge

Quanta - In physics quanta is the smallest amount a quantity can physically have, and all quantities are made up of multiples of this "quanta". Fundamental Charge: $q_e = 1.6 \times 10^{-19}$ Coulombs is the smallest possible charge, charge is quantized. Integer number of indivisible charges. All charge is a multiple of q_e. $1q_e$, $2q_e$, $3q_e$, $4q_e$, … .

Total charge on an object: $\mathbf{Q = Nq_e}$ where N is # of excess or missing electrons

Example: How many extra electrons must a balloon have if the net charge on the balloon is negative 12 μC?

$Q = Nq_e$ therefore $N = Q/q_e = 12 \times 10^{-6}\ C / 1.6 \times 10^{-19}\ C = 7.5 \times 10^{13}$ electrons

<u>Charge mobility</u> – Charge is able to move and redistribute on an object and to move to other objects.

- If the object is a conductor (like metal) and this object has a net charge, then the charge is free to move and distributes itself over the entire surface of the conducing object. If the conductor is discharged, all the charge over the entire surface will discharge.

- If the object is an insulator (plastic) and this object has a net charge, then the charge does not move freely, it stays where it was placed or created and does not move around the object. If the object is discharged only the surface that is touched discharges and the rest of the object will stay charged.

Example: If two identical conducing spheres each with a different net charge sit on an insulated base, are brought together and make contact, charge will move from one to the other due to the electrostatic force. By convention the positive charge flows to the negative charge. The charge will flow until an equilibrium is reached and both spheres have the same charge.

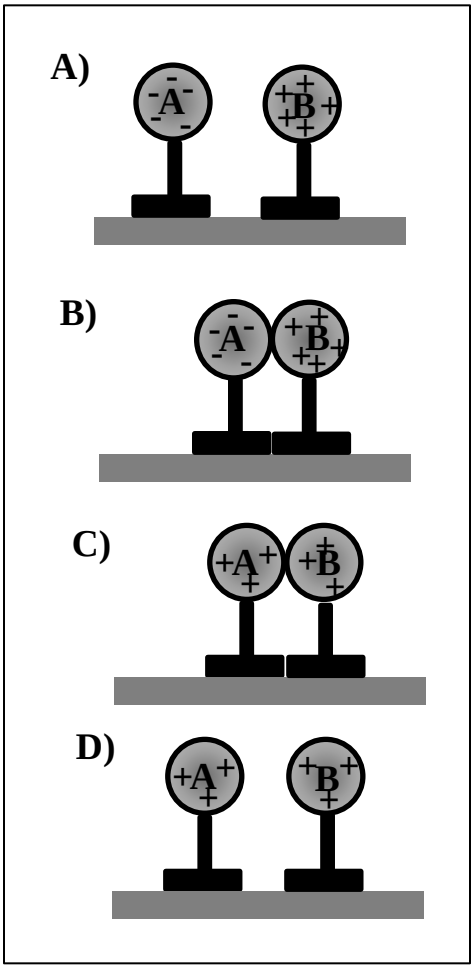
A) Sphere A has a 3 μC negative charge and Sphere B has a 4 μC positive charge when separated.

B) When the two spheres touch, the positive charges on Sphere B are attracted to the negative charges on sphere A and vice versus.

C) It takes 3 μC of positive charge from Sphere B to neutralize the 3 μC of negative charge from Sphere A, therefore 3 μC of positive charge moves from Sphere B to Sphere A.

   This leaves Sphere B with 1 μC of positive charge and Sphere A neutral, the 1 μC of positive charge will spread out over both spheres, this would result in Sphere B having ½ μC of positive charge and Sphere A having ½ μC of positive charge for a total of 1 μC of positive charge.

D) When the spheres are separated, each sphere will have a net charge of ½ μC of positive charge. Therefore, the total positive charge transferred from Sphere B to Sphere A is Q_B = 3 μC + ½ μC = 3 ½ μC of positive charge.

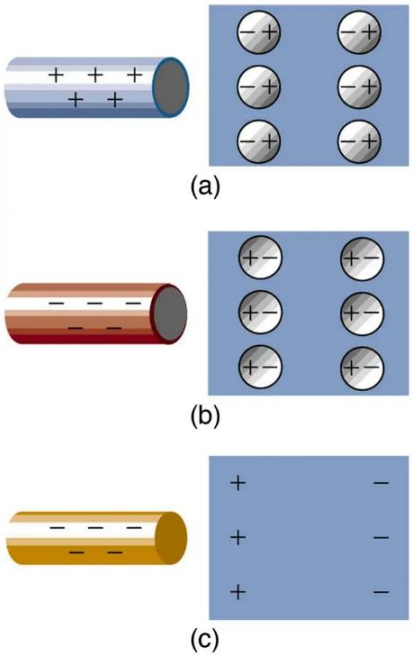


Induced Charge - A charge separation created in a region on an object by attraction of the opposite type of charge and repulsion by the same type of charge due to an external net charge. The charge separation in the object is influenced by an external charge and can rearrange and be redistributed on the object.

Polarization – is the relative shift of charge in an object due to an external charge near but not touching the object. The charge in the object that is the same type as that of the external net charge will be moved away from the external charge and the opposite charge from the external charge will move toward the external charge resulting in a charge separation with alignment of the charge pairs inside the neutral object.

(a) If there is a positively charged plastic rod is brought near but not touching a neutral insulator, the electrons in that neutral object will be drawn towards the positively charged rod and the positive protons in the plastic will be pushed away from the positively charged on the rod. The electrons and protons in the atom will be separated or polarized.

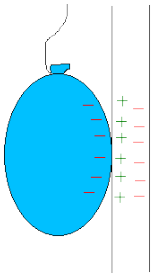
(b) The opposite is happening for a negatively charged rod. If there is a negatively charged plastic rod is brought near but not touching a neutral insulator, the electrons in that neutral object will be pushed away from the negative charged rod and the positive protons in the plastic will be drawn towards the negative charge on the rod. The electrons and protons in the atom will be separated or polarized.



(c) For a conductor the charge is free to move therefore the electrons in the conductor would be pushed away from the negatively charged rod creating a positively charged surface near the charged rod.

- Polarization induces charge separation exposing the separate charges that would under normal circumstances be neutral, they act as if there is a net charge. This is the reason that a charged balloon sticks a neutral wall, or a charged rod can pick up neutral pieces of paper

Temporary Induction of Charge - If there is a neutral wall and a negative charged balloon is brought near the wall, the charge is displacement in the neutral wall. The negative balloon attracts the positive charge near the walls surface and the negative charge in the wall is moved away from the negative balloon. This is an induced charge separation on the wall, the charge in the wall is polarized. The balloon will "stick" to the wall due to the electrostatic attraction on the negative (balloon) and positive (induced on wall) charges. When the charged balloon is removed the charges in the wall return to their pervious neutral positions.



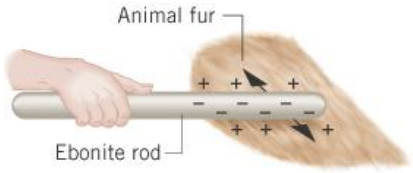
- Neutral objects can be <u>attracted</u> to charged objects (any type of charge) by polarization but <u>never repelled</u> by a net charge.

<u>Grounding</u> – An object that is large enough that it can take a small excess charge without itself becoming significantly charged. Common examples are the human body and the earth itself.

<u>A few Methods For Producing a Net Charge</u> – there are three methods for creating a net charge on an object, by rubbing the object thereby removing or adding charge, by physical contact thereby transferring charge or by electrostatic induction.

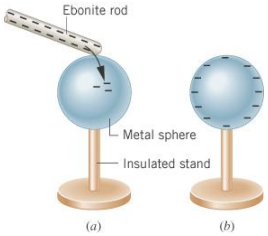
1) **<u>Electrification by Rubbing (Triboelectric Charging)</u> -** When two materials are rubbed together, the materials can either give up or attract electron charge, the materials create an equal and opposite charge on each of the materials being rubbed. Charge can be positive or negative and is dependent on how strongly the material holds on to or gives up its electrons.

   To produce a Positive Charge - Fur, glass, hair, nylon and silk become positively charged (+), therefore they losses electrons.



   To produce a Negative Charge - Teflon, rubber, amber, wood and cotton become negatively charged (-) therefore they gain electrons.

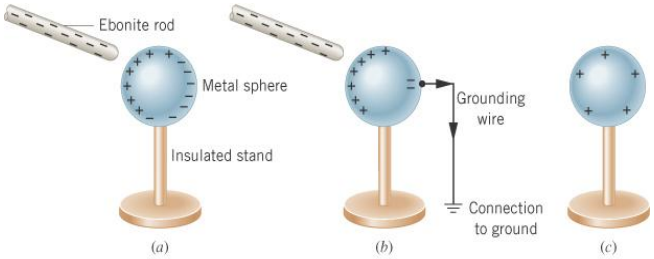
2) **<u>Direct Contact</u>** – Charge is transferred by physical contact with a charged object. Because each individual charge on the charged rod repels each other, when the charge has a chance to move farther away, like moving to a metal sphere, then they will transfer to the object in contact.



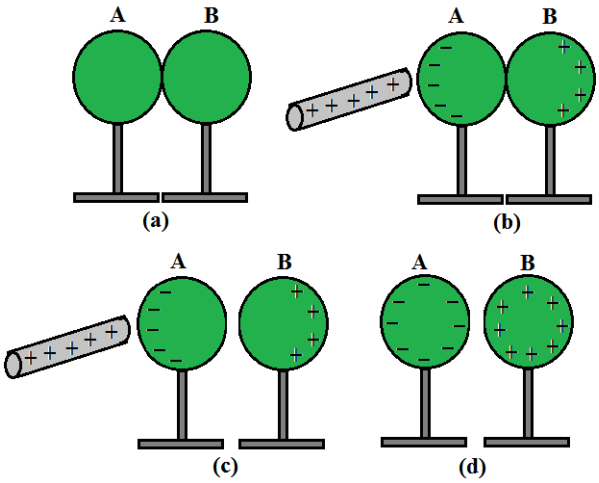
3) **<u>Charging by Induction</u>** – Using the polarization effect a movement of charge can be induced in an object due to an external charge, this can produce a temporary or an apparent net charge on an objects surface.

<u>Charging a conductor by induction</u> - When a negatively charged object (like the Ebonite rod) is brought near a neutral metal conducting sphere but does not touch, the electrons in the metal sphere are replied by the negative charge on the rod and move to the other side of the sphere. If the sphere is grounded, meaning the electrons are given a path to move away from the negatively charged rod then they will move to ground. If the path to ground is remove not letting the charge back on the sphere and then the rod is

moved away the result is a metal sphere with a net positive charge. The same can be done with a positively charged rod and the net effect would be to produce a net negative charge on the sphere.



If two conducting spheres (A and B) are in contact and a positively charged object is brought near sphere A but does not touch, then the negative charge will be attracted to the positive external charge and the positive charge in the spheres will be repelled by the external charge. This will leave more negative charge on sphere A and more positive charge on sphere B. If the two spheres are then separated, sphere A will have a net negative charge and sphere B will have a net positive charge.



Examples and applications of Static Charge

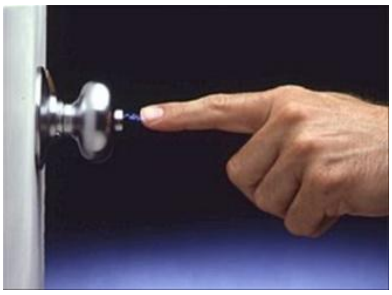
- Static Cling – The net charge is created by rubbing of clothes, equal amount of positive and negative charge on each piece of clothing. The dryer can sense this and know when the clothes are dry.



- Wrapping Tape – The net charge created by pulling tape apart, makes tape bend and curl due to the electrostatic forces from the net charges created on each piece of tape.



- Shock by doorknob – The net charge created by friction of shoes on a rug charges (Triboelectric charging) your body and then discharges on doorknob or another conductor such as a friend. (Ground)



- Electrostatic air filters - use a charged grid to create a net charge on the particles in the air, these charged particles are then attracted to an oppositely charged collection grid thereby producing particle free air.

